

Linear Quadratic Regulator

Consider the following optimal control problem: Linear Quadratic Regulator

$$\begin{aligned} \min_{\mathbf{u}} \quad & \sum_{k=0}^{N-1} \mathbf{x}_k^T Q_k \mathbf{x}_k + \mathbf{u}_k^T R_k \mathbf{u}_k + \mathbf{x}_N^T Q_N \mathbf{x}_N \\ \text{s.t.} \quad & \mathbf{x}_{k+1} = A_k \mathbf{x}_k + B_k \mathbf{u}_k \\ & \mathbf{x}_0 = \mathbf{x}_{\text{current}} \end{aligned} \quad Q \text{ is PSD, } R \text{ is PD}$$

- **Linear** dynamics
- **Quadratic** cost
- **Regulates** state and control to zero
- This is a quadratic program. Can solve control sequence given initial state.

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Can we find optimal policy $\mathbf{u}_k^* = \pi(\mathbf{x}_k)$? Optimal control for any current state, rather than solve a QP each time.

→ Apply dynamic programming!

LQR: Discrete time

$$\begin{aligned} \min_{\mathbf{u}} \quad & \sum_{k=0}^{N-1} \mathbf{x}_k^T Q_k \mathbf{x}_k + \mathbf{u}_k^T R_k \mathbf{u}_k + \mathbf{x}_N^T Q_N \mathbf{x}_N \\ \text{s.t.} \quad & \mathbf{x}_{k+1} = A_k \mathbf{x}_k + B_k \mathbf{u}_k \\ & \mathbf{x}_0 = \mathbf{x}_{\text{current}} \end{aligned}$$

Recall the Bellman equation

$$V^*(\mathbf{x}_k, t_k) = \min_{\mathbf{u}_k \in \mathcal{U}} g(\mathbf{x}_k, \mathbf{u}_k, t_k) + V^*(f(\mathbf{x}_k, \mathbf{u}_k, t_k), t_{k+1})$$

$$g(\mathbf{x}_k, \mathbf{u}_k, t_k) =$$

$$f(\mathbf{x}_k, \mathbf{u}_k, t_k) =$$

$$V^*(\mathbf{x}_N, t_N) =$$

Discrete-time finite horizon LQR (Time-invariant)

1. set $P_N = Q_N$

2. for $k = N, \dots, 1$,

$$P_{k-1} = Q + A^T P_k A - A^T P_k B (R + B^T P_k B)^{-1} B^T P_k A$$

$$K_{k-1} = -(R + B^T P_k B)^{-1} B^T P_k A$$

3. for $k = 0, \dots, N - 1$, $\mathbf{u}_k^* = K_k \mathbf{x}_k$

Discrete-time infinite horizon LQR (Time-invariant)

1. set $P_{k-1} = P_k = P_\infty$

2.

$$P_\infty = Q + A^T P_\infty A - A^T P_\infty B (R + B^T P_\infty B)^{-1} B^T P_\infty A$$

$$K_\infty = -(R + B^T P_\infty B)^{-1} B^T P_\infty A$$

Discrete-time finite horizon LQR (Time-varying)

What changes?

1. set $P_N = Q_N$
2. for $k = N, \dots, 1$,

$$P_{k-1} = Q + A^T P_k A - A^T P_k B (R + B^T P_k B)^{-1} B^T P_k A$$

$$K_{k-1} = -(R + B^T P_k B)^{-1} B^T P_k A$$

3. for $k = 0, \dots, N - 1$, $\mathbf{u}_k^* = K_k \mathbf{x}_k$

Tracking LQR

- Consider a tracking a *desired trajectory* $(\tilde{\mathbf{x}}_k, \tilde{\mathbf{u}}_k)$

$$(\mathbf{x}_k - \tilde{\mathbf{x}}_k)^T Q_k (\mathbf{x}_k - \tilde{\mathbf{x}}_k) + (\mathbf{u}_k - \tilde{\mathbf{u}}_k)^T R_k (\mathbf{u}_k - \tilde{\mathbf{u}}_k)$$

- Consider *error* states, controls, and dynamics
 - $\delta \mathbf{x}_k = \mathbf{x}_k - \tilde{\mathbf{x}}_k$
 - $\delta \mathbf{u}_k = \mathbf{u}_k - \tilde{\mathbf{u}}_k$
 - $\delta \dot{\mathbf{x}}_k = \dot{\mathbf{x}}_k - \dot{\tilde{\mathbf{x}}}_k = A_k \mathbf{x}_k + B_k \mathbf{u}_k - (A_k \tilde{\mathbf{x}}_k + B_k \tilde{\mathbf{u}}_k) = A_k \delta \mathbf{x}_k + B_k \delta \mathbf{u}_k$
- Want error $\rightarrow 0$
 - Apply standard LQR on *error* states, controls, and dynamics
 - $\delta \mathbf{u}_k^* = K_k \delta \mathbf{x}_k \Rightarrow \mathbf{u}_t^* = K_k \delta \mathbf{x}_k + \tilde{\mathbf{u}}_k$